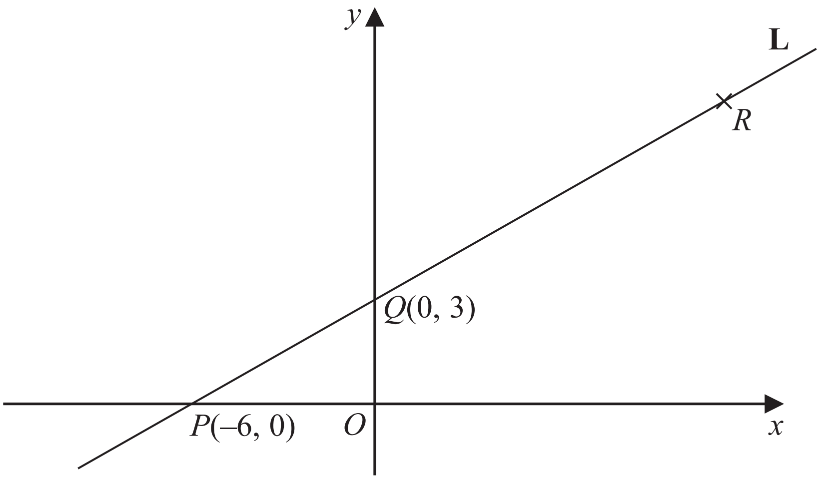
1 Here is a sketch of the line **L**.



The points $P(-6, 0)$ and $Q(0, 3)$ are points on the line **L**.

The point R is such that PQR is a straight line and $PQ : QR = 2 : 3$

(a) Find the coordinates of R.

(.......................... , ...)

(2)

(b) Find an equation of the line that is perpendicular to **L** and passes through Q.

....

(3)

(Total for Question 1 is 5 marks)

2 The points L, M and N are such that LMN is a straight line.

The coordinates of L are $(-3, 1)$

The coordinates of M are $(4, 9)$

Given that $LM : MN = 2 : 3$,

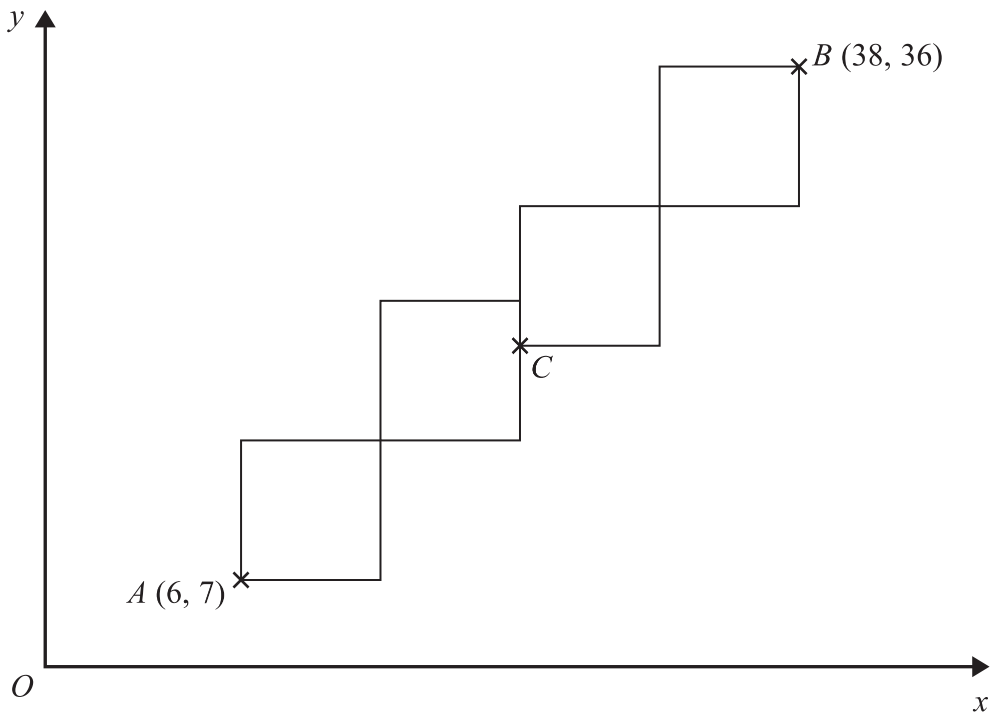
find the coordinates of N.

(.............. , ...)

(Total for Question 2 is 4 marks)

3 A pattern is made from four identical squares.

The sides of the squares are parallel to the axes.



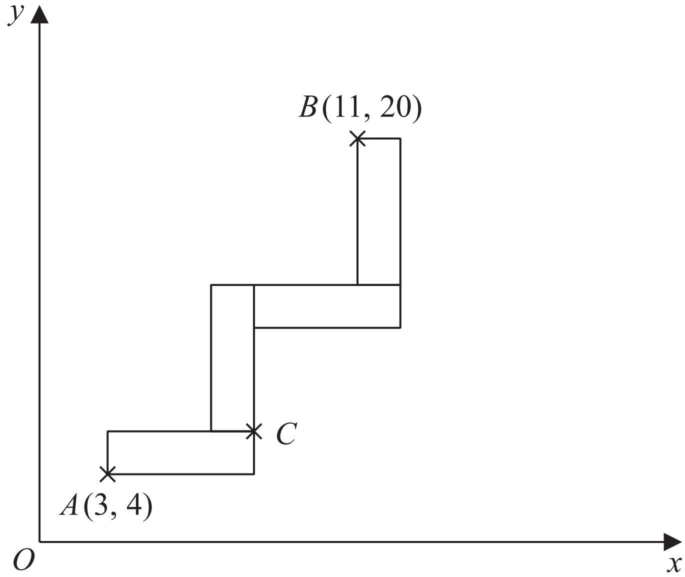
Point A has coordinates (6, 7)
Point B has coordinates (38, 36)
Point C is marked on the diagram.

Work out the coordinates of C.

(.............................. , ...)

(Total for Question 3 is 5 marks)

4 A pattern is made from four identical rectangles.
The sides of the rectangles are parallel to the axes.



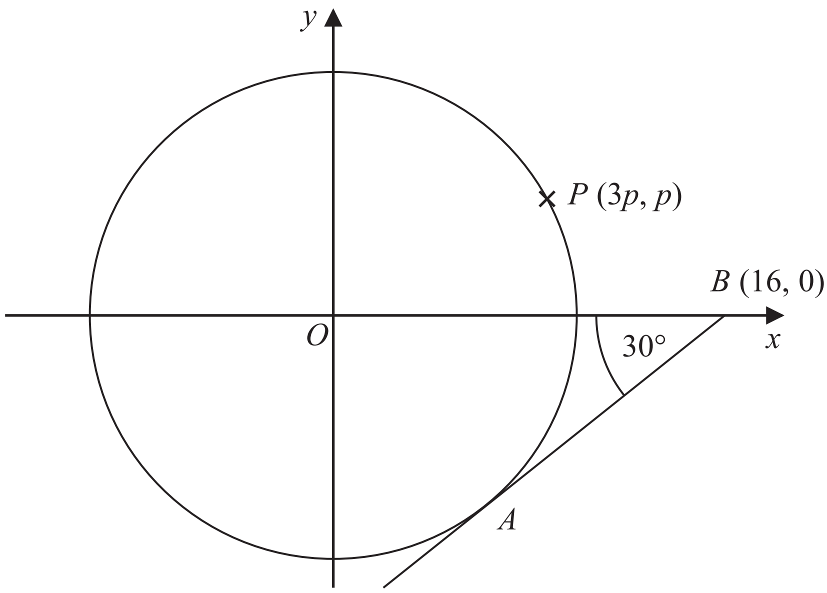
Point A has coordinates (3, 4)
Point B has coordinates (11, 20)
Point C is marked on the diagram.

Work out the coordinates of C.
You must show all your working.

(.............................. , ...)

(Total for Question 4 is 5 marks)

5 The diagram shows a circle, centre O.



AB is the tangent to the circle at the point A.
Angle $OBA = 30°$

Point B has coordinates $(16, 0)$
Point P has coordinates $(3p, p)$

Find the value of p.
Give your answer correct to 1 decimal place.
You must show all your working.

$p =$ ....

(Total for Question 5 is 4 marks)